

Scenario B Explanation — Conflict Between Burst Time and Priority:

Scenario B is designed to highlight the conflict between burst time and priority. It includes a long-burst, high-priority process (P1) alongside several short-burst, lower-priority processes (P2, P3, P4) and one medium-burst process (P5). This setup reveals a meaningful difference between the two algorithms: SJF will skip the long high-priority job in favor of shorter ones, while Priority Scheduling executes the high-priority job first regardless of its length.

Process	Arrival Time	Burst Time	Priority
P1	0	10	5
P2	0	2	1 (highest)
P3	0	2	1 (highest)
P4	0	2	1 (highest)
P5	0	3	2

SJF Execution Order: P1 → P2 → P3 → P4 → P5 (0–10 → 10–12 → 12–14 → 14–16 → 16–19)

Priority Execution Order: P1 → P2 → P3 → P4 → P5 (0–10 → 10–12 → 12–14 → 14–16 → 16–19)

SJF Average: WT = 8.40 | TAT = 12.20 | RT = 8.40

Priority Average: WT = 8.40 | TAT = 12.20 | RT = 8.40

Key Insight: In this particular case the Gantt charts match because all short-burst processes (P2, P3, P4) share the same top priority and all arrived at time 0, so both algorithms agree on the tie-breaking order. The important teaching point is that SJF chose them because they are short, while Priority chose them because of their priority number — the motivation differs even when the outcome coincides. In general, a workload where arrival times differ will produce visibly different Gantt charts for these two algorithms.